

1.3.19:

Solution:

Proof. ($\Rightarrow$) Assume that $W_1 \cup W_2$ is a subspace of V . We must show that either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. By way of contradiction we will suppose that $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$ so that $\exists v \in W_1, v \notin W_2$ and $u \in W_2, u \notin W_1$. Since $W_1 \cup W_2$ is a subspace it is closed under addition and $u + v \in W_1 \cup W_2$ and thus either $u + v \in W_1$ or $u + v \in W_2$. WLOG take $u + v \in W_1$ so that its additive inverse $-(u + v) \in W_1$. Since W_1 is also a subspace itself, it is also closed under vector addition and so $u + (-(u + v)) \in W_1 \Rightarrow u - u + v \in W_1 \Rightarrow v \in W_1$ which is a contradiction. An identical argument exists if $u + v \in W_2$. It follows that either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

($\Leftarrow$) Assume that $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$ for W_1 and W_2 subspaces of V . It follows that $W_1 \cup W_2 = W_1$ or $W_1 \cup W_2 = W_2$, either way the union is a subspace by assumption and we are done. $\square$

1.4.4a:

Solution: Consider the matrix form

$$\left(\begin{array}{cc|c} 1 & 1 & 1 \\ 2 & 3 & 0 \\ -1 & 0 & -3 \\ 1 & -1 & 5 \end{array} \right) \xrightarrow[r_4 \rightarrow r_4 - r_1]{r_2 \rightarrow r_2 - 2r_1, r_3 \rightarrow r_3 + r_1} \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & -2 & 4 \end{array} \right) \xrightarrow{r_2 \rightarrow -\frac{1}{2}r_2} \left(\begin{array}{cc|c} 1 & 1 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right) \Rightarrow X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}.$$

Thus, the system is consistent and the first polynomial is a linear combination of the others.

1.4.5g:

Solution: Consider the matrix form (with the 2×2 matrix represented as column vectors):

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ -1 & 0 & 0 & -3 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 0 & 4 \end{array} \right) \xrightarrow{r_2 \rightarrow r_2 + r_1} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 1 & 1 & 2 \\ 0 & 1 & 0 & 4 \end{array} \right) \xrightarrow{r_2 \leftrightarrow r_3} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 1 & 0 & 4 \end{array} \right)$$

$$\xrightarrow{r_4 \rightarrow r_4 - r_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -1 & 2 \end{array} \right)$$

$$\xrightarrow{r_4 \rightarrow r_4 + r_3} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right) \Rightarrow X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ -2 \\ 0 \end{pmatrix}$$

Thus the system is consistent and so the matrix $\begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix}$ is a linear combination of the other matrices and is thus included in their span.

1.4.6:

Solution: Consider

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{r_2 \rightarrow r_2 - r_1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{r_3 \rightarrow r_3 + r_2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

thus $\text{rank}(A) = \dim(\mathbb{F}^3) = 3$ and hence the 3 vectors generate $\mathbb{F}^3$.

1.5.2d:

Solution: Consider

$$A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 2 & 3 \\ -1 & 0 & 2 \\ 0 & 4 & 6 \end{pmatrix} \xrightarrow{r_3 \rightarrow r_3 + r_1} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 4 & 6 \end{pmatrix} \xrightarrow{r_4 \rightarrow r_4 - 2r_2} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \Rightarrow X = x_3 \begin{pmatrix} 2 \\ \frac{3}{2} \\ 1 \end{pmatrix}$$

for x_3 free. Since $\text{rank}(A) = 2 < 3$ the set is linearly dependent because there are infinitely many solutions to the homogenous system.

1.5.6:

Solution: Suppose scalars $a_{ij} \in F$ (for $1 \leq i \leq m$ and $1 \leq j \leq n$) satisfy

$$\sum_{i=1}^m \sum_{j=1}^n a_{ij} E^{ij} = O,$$

where O is the $m \times n$ zero matrix. We must show that every $a_{ij} = 0$.

Fix any k and l with $1 \leq k \leq m$ and $1 \leq l \leq n$. By the definition of E^{ij} , its (k, l) entry is

$$(E^{ij})_{kl} = \begin{cases} 1 & \text{if } (i, j) = (k, l), \\ 0 & \text{otherwise.} \end{cases}$$

Since matrix addition and scalar multiplication act entrywise, taking the (k, l) entry of both sides of the equation above gives

$$\sum_{i=1}^m \sum_{j=1}^n a_{ij} (E^{ij})_{kl} = 0.$$

Every term in this double sum vanishes except the one with $(i, j) = (k, l)$, which contributes $a_{kl} \cdot 1$. Hence the left-hand side equals a_{kl} , and so

$$a_{kl} = 0.$$

Because k and l were arbitrary, every coefficient a_{ij} is zero. Thus the only linear combination of the E^{ij} equal to O is the trivial one, and the set is linearly independent.

⁰LaTeX code for this document can be found on github [here](#)